

The Capital Asset Pricing Model (CAPM): An Introduction

The Capital Asset Pricing Model (CAPM) is one of the most important theories in modern finance. Developed in the 1960s by William Sharpe, John Lintner and others, CAPM helps us answer a fundamental question: What return should investors expect for taking on risk?

Think about it this way: if you're going to invest your money in something risky rather than keeping it in a safe government bond, you'll want to be compensated for that extra risk. But how much compensation is fair? CAPM gives us a framework to calculate this.

Types of Risk

Before CAPM, people understood that riskier investments should offer higher returns, but they didn't distinguish between different types of risk. CAPM's breakthrough insight is that there are two kinds of risk:

1. **Systematic Risk (Market Risk):** Risk that affects the entire market - things like economic recessions, inflation, or political instability. You *can't* eliminate this risk through diversification because it impacts everything.
2. **Unsystematic Risk (Specific Risk):** Risk specific to individual companies or industries - like a CEO scandal, a product recall, or new competition. This risk *can* be eliminated through diversification.

The key insight of the CAPM is that investors should only be compensated for systematic risk, because they can eliminate unsystematic risk for free by diversifying their portfolio. (Remember this is an important perspective because we want to determine how much return the firm needs to offer the capital market (investors) to obtain capital need for planned investment projects.)

Beta (β)

Beta (β) is CAPM's measure of systematic risk. It tells us: To what extent does a security's return move together with the overall market return?

- (1) $\beta = 1$: The security moves exactly with the market. If the market goes up 10%, this security typically goes up 10%.
- (2) $\beta > 1$: The security is more volatile than the market. If $\beta = 1.5$ and the market goes up 10%, this security typically goes up 15%.
- (3) $\beta < 1$: The security is less volatile than the market. If $\beta = 0.5$ and the market goes up 10%, this security typically goes up 5%.

The Efficient Frontier and Capital Market Line

Looking at the efficient frontier diagram, you can see how different companies have different risk-return-profiles. IBM has a high risk, while companies like Anheuser-Busch exhibit lower volatility of returns.

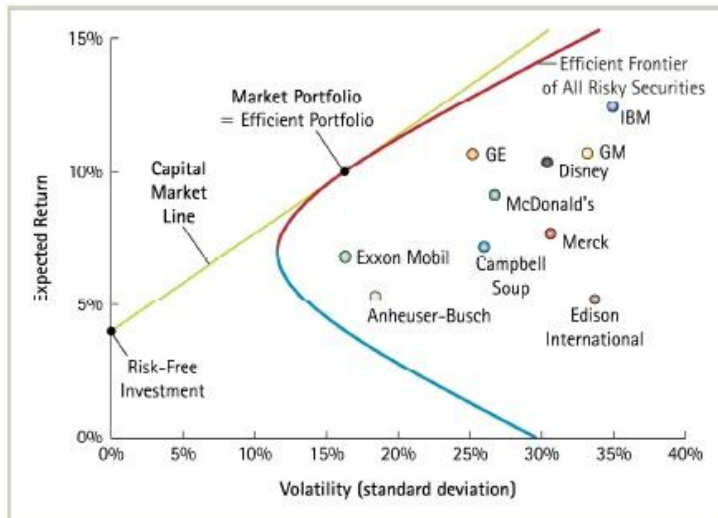


Figure 1 Efficient Frontier (Source: Berk/DeMarzo (2013)).

More importantly, the efficient frontier diagram reveals a crucial insight about optimal investing. The curved red line represents the efficient frontier - all possible combinations of risky assets that offer the highest expected return for each level of risk. However, when we introduce a risk-free investment (like government bonds), we can draw a straight line from the risk-free rate that just touches (is tangent to) the efficient frontier at one specific point - the market portfolio. This straight line is called the Capital Market Line (CML), and it represents the best possible risk-return combinations available to investors.

The key characteristic of the Capital Market Line is that it dominates the entire efficient frontier of risky assets alone. Any point on this line represents a portfolio combining the risk-free asset with the market portfolio in different proportions. Want lower risk? Hold more of the risk-free asset and less of the market portfolio. Want higher expected returns? Hold less (or even borrow to buy more) of the market portfolio. This is why the market portfolio - that single point where the line touches the efficient frontier - becomes the optimal risky portfolio for all investors, regardless of their risk preferences. Every rational investor should hold the same mix of risky assets (the market portfolio) and only vary their overall risk level by adjusting how much they combine it with the risk-free asset.

The CAPM Formula

CAPM gives us a simple but powerful equation to calculate the expected return on any investment:

$$r_i = r_f + \beta_i \times (E[r_{Mkt}] - r_f)$$

Let's break this down:

r_i = Expected return on security i (what we're trying to find)

r_f = Risk-free rate (usually government bond yield)

β_i = Beta of security i (its systematic risk)

$E[r_{Mkt}]$ = Expected return on the market portfolio

$E[r_{Mkt}] - r_f$ = Market risk premium (extra return for taking market risk)

A Practical Example

Let's say you're evaluating Microsoft stock:

- Risk-free rate (r_f) = 3%
- Expected market return ($E[r_{Mkt}]$) = 8%
- Microsoft's beta (β_i) = 1.2

Using CAPM:

$$r_i = 3\% + 1.2 \times (8\% - 3\%) = 3\% + 1.2 \times 5\% = 3\% + 6\% = 9\%$$

So according to CAPM, investors should expect a 9% return from Microsoft stock to be fairly compensated for its risk.

Real-World Applications

Although the CAPM is not perfect and has been extended over the years (in particular by Fama and French, two US scholars), it is used extensively in:

- (1) Corporate Finance: Calculating the cost of equity capital
- (2) Investment Analysis: Determining if stocks are fairly priced
- (3) Portfolio Management: Constructing efficient portfolios

CAPM is still influential because it provides a logical framework for thinking about risk and return. Its core insight - that only systematic risk deserves compensation - remains relevant today.